

Pierce Hoenigman

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EDUCATION

Cornell Tech, Cornell University New York, NY August 2025 – May 2026
Master of Engineering in Computer Science, Merit Scholarship GPA: 4.10/4.00
Relevant Courses: ML Engineering, Distributed Optimization for ML, Language Modeling, Generative Models, Reinforcement Learning, Entrepreneurial Finance

The University of Chicago Chicago, IL September 2021 – June 2025
Bachelor of Science in Biological Sciences, specialization in Quantitative Biology GPA: 3.99/4.00
Minors in Computer Science & Systems Bioengineering
Honors & Awards: Summa cum laude, Phi Beta Kappa, Biological Sciences Research Honors, Janet Rowley Scholar of the Biological Sciences, Robert Maynard Hutchins Scholar, Quad Scholar, Dean's List, DAAD RISE

TECHNICAL SKILLS

ML: Python (PyTorch, NumPy, pandas, matplotlib, biopython), R (Tidyverse, Bioconductor), C, CUDA
Full Stack: HTML, CSS, SQL, AWS (EC2, S3, IAM, VPC, RDS, SageMaker, Elastic Beanstalk)
Other: Linux/shell scripting, Slurm, Git, PitchBook

EXPERIENCE

Ranganathan Lab, Researcher Chicago, IL October 2023 – June 2025

- Designed interpretable protein generative models enabling property-tunable sequence generation
- Implemented and trained VAEs and other interpretability-focused models (RBMs, Hopfield networks) in PyTorch, integrating C-based optimizations and distributed training via Slurm
- Developed novel techniques for disentangling functionally relevant information from noise in protein evolutionary histories using information theory and custom model systems
- Achieved exponential gains in latent protein representation interpretability

Chicago Biomedical Consortium, Entrepreneurship Intern Chicago, IL July 2024 – September 2024

- Conducted structured market research and scientific literature mining to inform quantitative prioritization of Chicagoland biotech startup candidates
- Designed an automated data pipeline to aggregate biotech startup information and recent updates for 500+ companies by querying public APIs and merging heterogeneous datasets

SELECTED PROJECTS

GraphFormer for Spatial Proteomics December 2025 – January 2026

- Built and trained graph transformers on colorectal cancer spatial proteomics data to model intercellular signaling
- Quantified cell-type influence, spatial propagation range, and protein-specific neighbor effects

Pharma Transfer Learning November – December 2025

- Trained federated transfer learning models for physiological benchmark prediction tasks
- Compared against baseline learning methods to determine performance improvements and confirm feasibility of learning from heterogeneous siloed pharmaceutical data sources

Implicit Neural Pattern Representations February – March 2025

- Implemented coordinate-based neural networks to learn continuous image signal representations
- Significantly improved out-of-bounds generalization of image patterns using Kolmogorov-Arnold learned activations

ACTIVITIES

Quantitative Biosciences Center, Head Teaching Assistant Chicago, IL January 2023 – December 2025

- Designed and administered bioinformatics labs in R & Python for six courses using quantitative and statistical skills
- Lectured on programming fundamentals and helped 600+ students with coding, statistics, and ML questions

Genehackers (UChicago iGEM), Finance Director Chicago, IL October 2022 – June 2025

- Secured \$40,000 annually through corporate sponsorships/grant writing and optimized lab budget
- Managed a student synthetic biology research team developing a schistosomiasis vaccine, now in NIH testing
- Organized six guest speaker and networking events with scientific leaders and entrepreneurs in biotech and pharma